

Computer fundamental

UNIT: 1 INTRODUCTION TO COMPUTER

1. WHAT IS COMPUTER?

- The word “computer” is comes from the word “TO COMPUTE” means to calculate.
- A computer is normally considered to be a calculation device which can perform the arithmetic operations very speedily.
- A computer may be defined as a device which operates upon the data.
- Data can be in the form of numbers, letters, symbols, size etc. And it comes in various shapes & sizes depending upon the type of computer application.
- A computer can store, process & retrieve data as and when we desired.
- The fact that computer process data is so fundamental that many people have started calling as “Data Processor”.
- A computer first it gets the Data, does Process on it and then produces Information.



- DEFINATION OF COMPUTER
 - A computer is an electronic device which takes input from the user, processes it and gives the output as per user’s requirement.
 - So the main tasks of performed by the computer are:
 - Input
 - Process
 - Output

2. WRITE DOWN THE CHARACTERISTICS OF COMPUTER

Some important characteristics of the computer are as follow:

- Automatic:
 - Computers are automatic machines because it works by itself without human intervention.
 - Once it started on a job they carry on until the job is finished.
 - Computer cannot start themselves.

- They can work from the instructions which are stored inside the system in the form of programs which specify how a particular job is to be done.
- **Accuracy:**
 - The accuracy of a computer is very high.
 - The degree of accuracy of a particular computer depends upon its design.
 - Errors can occur by the computer. But these are due to human weakness, due to incorrect data, but not due to the technological weakness.
- **Speed:**
 - Computer is a very fast device. It can perform the amount of work in few seconds for which a human can take an entire year.
 - While talking about computer speed we do not talk in terms of seconds and milliseconds but in microseconds.
 - A powerful computer is capable of performing several billion (10⁹) simple arithmetic operations per second.
- **Diligence:**
 - Unlike human beings, a computer is free from monotony, tiredness & lack of concentration.
 - It can continuously work for hours without creating any error & without grumbling.
 - If you give ten million calculations to be performed, it will perform with exactly the same accuracy & speed as the first one.
- **Versatility:**
 - It is one of the most wonderful features about the computer.
 - One moment it is preparing the results of a particular examination, the next moment it is busy with preparing electricity bills and in between it may be helping an office secretary to trace an important letter in seconds.
- **Power of remembering:**
 - Computer can store and recall any amount of data because of its high storage capacity of its storage devices.
 - Every piece of information can be retained as long as desired by the user and can be recalled as and when required.
 - Even after several years, if the information is recalled, it will be as accurate as on the day when it was filled to the computers.
- **No I.Q.**
 - A computer is not a magical device; it processes no intelligence of its own.
 - Its I.Q. is zero.

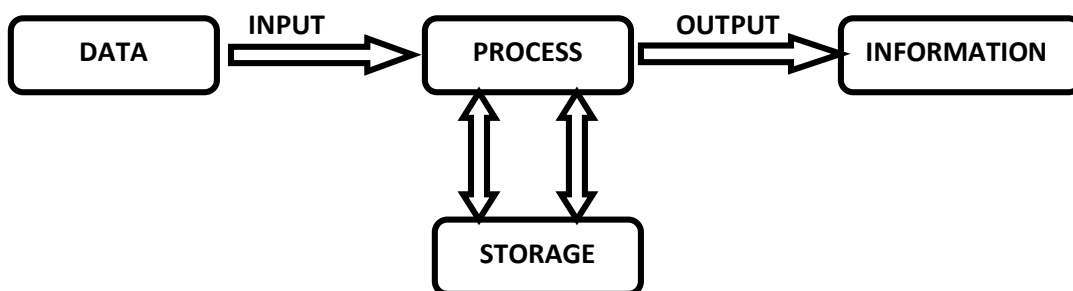
- It has to be told what to do & in what sequence.
- It cannot take its own decision.
- **No Fallings:**
 - A Computer has no feelings because they are machines.
 - Based on our feelings, task, knowledge and experience we often make certain judgments in our day today life.
 - But Computer goes exactly the way which we have given the instructions.

3. EXPLAIN THE DATA PROCESSING CYCLE OF COMPUTER.

- The computer Data Processing is any process that a computer program does to enter data & summarise, analyse or convert data into useable information.
- The process may be automated & run on a computer.
- It involves recording, analysing, storing, summarising & storing data.
- Because data are most useful when it is well presented & informative.

The Data Processing Cycle:

- Data Processing cycle described all activities which are common to all data processing systems from manual to electronic systems.
- These activities can be grouped in four functional categories, viz., data input, data processing, data output and storage, constituting what is known as a data processing cycle.
- The main aim of data processing cycle is to convert the data into meaningful information.
- Data processing system are often referred to as Information System.
- The Information System typically take raw Data as Input to produce Information as Output.



- The data processing cycle contains main four functions:
 - Data input

- Data process
- Data storage
- Data output
- **DATA INPUT**
 - The term input refers to the activities required to record data.
 - It's a process to entered data in to computer system.
 - So before we input any data, it is necessary to check or verify the data context.
- **DATA PROCESSING**
 - The term processing includes the activities like classifying, storing, calculating, comparing or summarising the data.
 - The processing means to use techniques to convert the data into meaningful information.
- **DATA OUTPUT**
 - It's a communication function which transmits the information to the outside world.
 - After completed the process the data are converted into the meaningful in
 - Sometimes the output also includes the decoding activity which converts the electronically generated information into human readable form.
- **DATA STORAGE**
 - It involves the filling of data & information for future use.

4. EXPLAIN THE CLASSIFICATION OF THE COMPUTER BY DATA PROCESSED

The computers are divided mainly three types on the based on data processed:

1. Analog computers
2. Digital computers
3. Hybrid computers

Analog computers:

- In Analog Computers, data is represented as continuously varying voltage and operate essentially by measuring rather counting.
- As the data is continuously variable, the results obtained are estimated and not exactly repeatable.
- It can able to perform multiple tasks simultaneously and also capable to work effectively with the irrational number. E.g. $1/8 = 0.125$ and $1/6=0.1666$

- Voltage, temperature and pressure are measured using analog devices like voltmeters, thermometers and barometers.

Digital Computers

- The digit computer is a machine based on digital technology which represents information by numerical digit.
- In Digital Computers data is represented as discrete units of electrical pulses. The data is measured in quantities represented as either the 'on' or 'off' state.
- Therefore, the results obtained from a digital computer are accurate.
- Virtually all of today's computers are based on digital computers.

Hybrid Computers

- It combines the good features of both analog & digital computers.
- It has a speed of analog computer & accuracy of digital computer.
- Hybrid Computers accept data in analog form and present output also in digitally.
- The data however is processed digitally.
- Therefore, hybrid computers require analog-to-digital and digital-to-analog converters for output.

5. EXPLAIN THE CLASSIFICATION OF THE COMPUTER BY DATA PROCESSING:

The computers are classified in four types on the based on data processing.

- Micro computer
- Mini computer
- Mainframe computer
- Super computer

Micro Computer:

- Micro computers are the computers with having a microprocessor chip as it central processing unit.
- Originated in late 1970s.
- First micro computer was built with 8 bit processor.
- Microcomputer is known as personal computer.
- Designed to use by individual whether in the form of pc's, workstation or notebook computers.
- Small in size and affordable for general people.
- Ex: IBM PC, IBM PC/XT, IBM PC/AT

Micro Computer:

- Mini computers are originated in 1960s.
- Small mainframes that perform limited tasks.
- Less expensive than mainframe computer.
- Mini computers are Lower mainframe in the terms of processing capabilities.
- Capable of supporting 10 to 100 users simultaneously.
- In 1970s it contains 8 bit or 12 bit processor.
- Gradually the architecture requirement is grown and 16 and 32 bit.
- Minicomputers are invented which are known as supermini computers.
- Ex: IBM AS400

Mainframe Computer:

- A very powerful computer which capable of supporting thousands of user simultaneously.
- It contains powerful data processing system.
- It is capable to run multiple operating systems.
- It is capable to process 100 million instructions per second.
- Mainframes are very large & expensive computers with having larger internal storage capacity & high processing speed.
- Mainframes are used in the organization that need to process large number of transaction online & required a computer system having massive storage & processing capabilities.
- Mainly used to handle bulk of data & information for processing.
- Mainframe system is housed in a central location with several user terminal connected to it.
- Much bigger in size & needs a large rooms with closely humidity & temperature.
- IBM & DEC are major vendors of mainframes.
- Ex : MEDHA, SPERRY, IBM, DEC, HP, HCL

Super Computer:

- Most powerful & most expensive computer.
- Used for complex scientific application that requires huge processing power.
- Used multiprocessor technology to perform the calculation very speedy.
- They are special purpose computers that are designed to perform some specific task.

- The cost of the super computer is depended on its processing capabilities & configuration.
- The speed of modern computer is measured in gigaflops, teraflops and petaflops.
 - Gigaflops= 10^9 arithmetic operation per second.
 - Teraflops= 10^{12} arithmetic operation per second.
 - Petaflops= 10^{15} arithmetic operation per second.
- Ex: PARAM , EKA, BLUE GENE/P

6. EXPLAIN THE GENERATION OF THE COMPUTERS.

In Computer language, “Generation” is a set of Technology. It provides a framework for the growth of the computer technology. There are totally Five Computer Generations till today. Discussed as following.

First Generation:

- Duration: 1942-1955
- Technology: vacuum tube
 - Used as a calculating device.
 - Performed calculations in milliseconds.
 - To bulky in size & complex design.
 - Required large room to place it.
 - Generates too much heat & burnt.
 - Required continuously hardware maintenance.
 - Generates much heat so must air-conditioner rooms are required.
 - Commercial production is difficult & costly.
 - Difficult to configure.
 - Limited commercial use.
 - ENIAC, EDVAC, EDSAC are example of 1st generation computer.

Second Generation:

- Duration: 1955-1964
- Technology: transistor
 - 10 times Smaller in size than 1st generation system.
 - Less heat than 1st generation computers.
 - Consumed less power than 1st generation system.
 - Computers were done calculations in microseconds.
 - Air-conditioner is also required.
 - Easy to configure than 1st generation computers.

- More reliable in information.
- Wider commercial use.
- Large & fast primary/secondary storage than 1st generation computers.

Third Generation:

- Duration: 1965-1975
- Technology: IC chip
 - Smaller in size than 1st & 2nd generation computers.
 - Perform more fast calculations than 2nd generation systems.
 - Large & fast primary/secondary storage than 2nd generation computers.
 - Air –conditioner is required.
 - Widely used for commercial applications.
 - General purpose computers.
 - High level languages like COBOL & FORTAN are allowed to write programs.
 - Generate less heat & consumed less power than 2nd generation computer.

Fourth Generation:

- Duration: 1975-1989
- Technology: Microprocessor chip
 - Based on LSI & VLSI microprocessor chip.
 - Smaller in size.
 - Much faster than previous generations.
 - Minimum hardware maintenance is required.
 - Very reliable as computer to previous generation computers.
 - Totally general purpose computer.
 - Easy to configure.
 - Possible to use network concept to connect the computer together.
 - NO requirement of air-conditioners.
 - Cheapest in price.

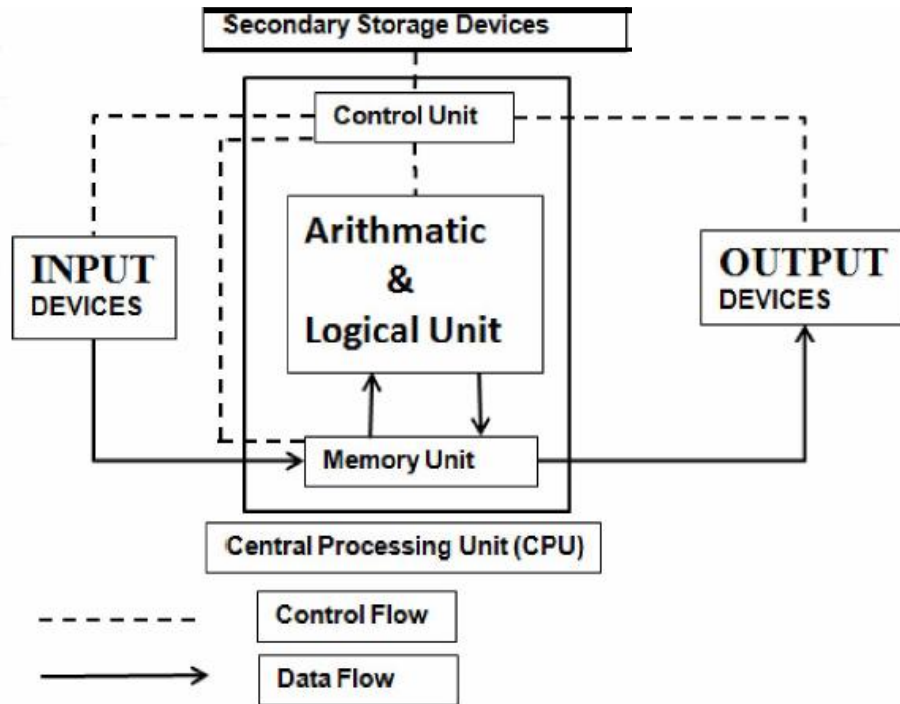
Fifth Generation:

- Duration: 1989 to Present
- Technology: ULSI microprocessor chip
 - Much smaller & handy.
 - Based on the ULSI chip which contains 100 million electronic components.
 - The speed of the operations is increased.
 - Consumed less power.
 - Air-conditioner is not required.
 - More user friendly interface with multi-media features.
 - High level languages are allowed to write programs.
 - Larger & faster primary/secondary storage than previous generations.

- Notebook computers are the example of 5th generation computers.

7. EXPLAIN THE BLOCK DIAGRAM OF COMPUTER OR EXPLAIN THE SIMPLE MODEL COMPUTER.

A simple computer system comprises the basic components like Input Devices, CPU (Central Processing Unit) and Output Devices as under:



- **Input Devices:**
 - The devices which are used to entered data in the computer systems are known as input devices.
 - Keyboard, mouse, scanner, mike, light pen etc are example of input devices.

FUNCTION OF INPUT DEVICES

- Accept the data from the outside worlds.
- Convert that data into computer coded information.
- Supply this data to CPU for further processing.

- **Output Devices:**
 - The devices which display the result generated by the computer are known as output devices.
 - Monitor, printer, plotter, speaker etc are the example of output devices.

FUNCTIONS OF OUTPUT DEVICES

- Accept the result form the CPU.
- Convert that result into human readable form.

- Display the result on the output device.
- **Memory Unit:**
 - The data & instruction have to store inside the computer before the actual processing start.
 - Same way the result of the computer must be stored before passed to the output devices. This tasks performed by memory unit.

FUNCTIONS OF MEMORY UNIT

- Store data & instruction received from input devices.
- Store the intermediate results generated by CPU.
- Store the final result generated by CPU.
- **Arithmetical & Logical Unit:**
 - The ALU is the place where actual data & instruction are processed.
 - All the calculations are performed & all comparisons are made in ALU.
 - Performs all arithmetical & logical operations.
 - An arithmetic operation contains basic operations like addition, subtraction, multiplication, division.
 - Logical operations contains comparison such as less than, greater than, less than equal to, greater than equal to, equal to, not equal to.
- **Control Unit:**
 - It controls the movement of data and program instructions into and out of the CPU, and to control the operations of the ALU.
 - In sort, its main function is to manage all the activities within the computer system.
 - Controls the internal parts as well as the external parts related with the computer.
- **CPU:**
 - The Unit where all the processing is done is called as Central Processing Unit.
 - It contains many other units under it.
 - Main of them are:- Control Unit And ALU (Arithmetic & Logic Unit)